

Commerce XI Maths App

Exercise 1.1

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 1, Unit 1 - Sets and Relations | Total questions: 34

Question 1 - Q1(i)

Describe the following sets in Roster form

$\{x/x \text{ is a letter of the word 'MARRIAGE'}\}$

Options

- A. $\{M, A, R, I, G, E\}$
- B. $\{M, A, R, I, G\}$
- C. $\{M, A, R, I, E\}$
- D. $\{M, A, R, I, G, E, S\}$

Correct answer: $\{M, A, R, I, G, E\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{M, A, R, I, G, E\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 2 - Q1(ii)

Describe the following sets in Roster form

$\{x/x \text{ is an integer, } -1/2 < x < 9/2\}$

Options

- A. $\{0, 1, 2, 3, 4\}$
- B. $\{-1, 0, 1, 2, 3, 4\}$
- C. $\{0, 1, 2, 3, 4, 5\}$
- D. $\{1, 2, 3, 4\}$

Correct answer: $\{0, 1, 2, 3, 4\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{0, 1, 2, 3, 4\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 3 - Q1(iii)

Describe the following sets in Roster form

$\{x/x = 2n, n \in \mathbb{N}\}$

Options

- A. $\{2, 4, 6, 8, \dots\}$
- B. $\{0, 2, 4, 6, \dots\}$

C. $\{1, 3, 5, 7, \dots\}$

D. $\{2, 4, 6, 8\}$

Correct answer: $\{2, 4, 6, 8, \dots\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{2, 4, 6, 8, \dots\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 4 - Q2(i)

Which set-builder form represents $\{0\}$?

(W is the set of whole numbers and N is the set of natural numbers.)

Options

A. $\{x / x \in W, x \notin N\}$

B. $\{x | x \in W, x = 1\}$

C. $\{x | x \in W, x < 2\}$

D. $\{x | x \in Z, x < 1\}$

Correct answer: $\{x / x \in W, x \notin N\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{x / x \in W, x \notin N\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 5 - Q2(ii)

Which set-builder form represents $\{0, \pm 1, \pm 2, \pm 3\}$?

Options

A. $\{x / -3 \leq x \leq 3, x \in Z\}$

B. $\{x | x \in Z, -3 < x < 3\}$

C. $\{x | x \in N, -3 \leq x \leq 3\}$

D. $\{x | x \in Z, -3 \leq x < 3\}$

Correct answer: $\{x / -3 \leq x \leq 3, x \in Z\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{x / -3 \leq x \leq 3, x \in Z\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 6 - Q2(iii)

Which set-builder form represents $\{1/2, 2/5, 3/10, 4/17, 5/26, 6/37, 7/50\}$?

Options

- A. $\{x / x = n/(n^2 + 1), n \in \mathbb{N}, n \leq 7\}$
- B. $\{x \mid x = n/(n^2 + 1), n \in \mathbb{N}, n < 7\}$
- C. $\{x \mid x = n/(n + 1), n \in \mathbb{N}, n \leq 7\}$
- D. $\{x \mid x = n^2/(n^2 + 1), n \in \mathbb{N}, n \leq 7\}$

Correct answer: $\{x / x = n/(n^2 + 1), n \in \mathbb{N}, n \leq 7\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{x / x = n/(n^2 + 1), n \in \mathbb{N}, n \leq 7\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 7 - Q3(i)

Let $A = \{x \mid 6x^2 + x - 15 = 0\}$, $B = \{x \mid 2x^2 - 5x - 3 = 0\}$, and $C = \{x \mid 2x^2 - x - 3 = 0\}$.

Find $A \cup B \cup C$.

Options

- A. $\{-5/3, -1, -1/2, 3/2, 3\}$
- B. $\{-5/3, 3/2, 3\}$
- C. $\{-5/3, -1/2, 3/2, 3\}$
- D. $\{-5/3, -1, -1/2, 3/2\}$

Correct answer: $\{-5/3, -1, -1/2, 3/2, 3\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{-5/3, -1, -1/2, 3/2, 3\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 8 - Q3(ii)

Let $A = \{x \mid 6x^2 + x - 15 = 0\}$, $B = \{x \mid 2x^2 - 5x - 3 = 0\}$, and $C = \{x \mid 2x^2 - x - 3 = 0\}$.

Find $A \cap B \cap C$.

Options

- A. $\{\}$
- B. $\{3/2\}$

C. $\{3\}$

D. $\{-1/2\}$

Correct answer: $\{ \}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{ \}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 9 - Q4

Let A, B, and C be the sets of distinct letters in “college”, “marriage”, and “luggage”.

What is $A - (B \cup C)$?

Options

A. $\{c, o\}$

B. $\{o\}$

C. $\{c, o, l\}$

D. $\{c, o, g\}$

Correct answer: $\{c, o\}$

Explanation and steps

Let M be Mathematics and P be Physics. From $n(M \cup P) = n(M) + n(P) - n(M \cap P)$: $20 = 12 + n(P) - 4$, hence $n(P) = 12$.

Question 10 - Q5(i)

Given $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8\}$.

Evaluate $A \cup (B \cap C)$.

Options

A. $\{1, 2, 3, 4, 5, 6\}$

B. $\{3, 4, 5, 6\}$

C. $\{1, 2, 3, 4, 5, 6, 7, 8\}$

D. $\{4, 5, 6\}$

Correct answer: $\{1, 2, 3, 4, 5, 6\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{1, 2, 3, 4, 5, 6\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 11 - Q5(ii)

Given $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8\}$.

Evaluate $A \cap (B \cup C)$.

Options

- A. $\{3, 4\}$
- B. $\{4\}$
- C. $\{3, 4, 5, 6\}$
- D. $\{1, 2, 3, 4\}$

Correct answer: $\{3, 4\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{3, 4\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 12 - Q5(iii)

Given $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8\}$.

Let the universal set be $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

Evaluate $(A \cup B)'$.

Options

- A. $\{7, 8, 9, 10\}$
- B. $\{1, 2, 3, 4, 5, 6\}$
- C. $\{5, 6, 7, 8, 9, 10\}$
- D. $\{7, 8\}$

Correct answer: $\{7, 8, 9, 10\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{7, 8, 9, 10\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 13 - Q5(iv)

Given $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8\}$.

Let the universal set be $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

Evaluate $(A \cap B)'$.

Options

- A. $\{1, 2, 5, 6, 7, 8, 9, 10\}$
- B. $\{3, 4\}$

C. {1, 2, 5, 6}

D. {5, 6, 7, 8, 9, 10}

Correct answer: {1, 2, 5, 6, 7, 8, 9, 10}

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives {1, 2, 5, 6, 7, 8, 9, 10}. The other choices omit a required element, include an invalid element, or use a different operation.

Question 14 - Q5(v)

Given $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8\}$.

Let the universal set be $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

Evaluate $(A \cap B) \cup (A \cap B')$.

Options

A. {1, 2, 3, 4}

B. {3, 4}

C. {1, 2}

D. {1, 2, 3, 4, 5, 6}

Correct answer: {1, 2, 3, 4}

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives {1, 2, 3, 4}. The other choices omit a required element, include an invalid element, or use a different operation.

Question 15 - Q5(vi)

Given $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8\}$.

Let the universal set be $X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

Evaluate $(A \cap B) \cup (A' \cap B)$.

Options

A. {3, 4, 5, 6}

B. {3, 4}

C. {5, 6}

D. {1, 2, 3, 4, 5, 6}

Correct answer: {3, 4, 5, 6}

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives {3, 4, 5, 6}. The other choices omit a required element, include an invalid element, or use a different operation.

Question 16 - Q5(vii)

Given $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8\}$.

Find $n(A \cup B)$.

Options

A. 6

B. 8

C. 4

D. 2

Correct answer: 6

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 6. The other choices omit a required element, include an invalid element, or use a different operation.

Question 17 - Q6(i)

If $n(X) = 50$, $n(A) = 35$, $n(B) = 20$, and $n(A' \cap B') = 5$.

Find $n(A \cup B)$.

Correct answer: 45

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 45. The other choices omit a required element, include an invalid element, or use a different operation.

Question 18 - Q6(ii)

If $n(X) = 50$, $n(A) = 35$, $n(B) = 20$, and $n(A' \cap B') = 5$.

Find $n(A \cap B)$.

Correct answer: 10

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 10. The other choices omit a required element, include an invalid element, or use a different operation.

Question 19 - Q6(iii)

If $n(X) = 50$, $n(A) = 35$, $n(B) = 20$, and $n(A' \cap B') = 5$.
Find $n(A' \cap B)$.

Correct answer: 10

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 10. The other choices omit a required element, include an invalid element, or use a different operation.

Question 20 - Q6(iv)

If $n(X) = 50$, $n(A) = 35$, $n(B) = 20$, and $n(A' \cap B') = 5$.
Find $n(A \cap B')$.

Correct answer: 25

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 25. The other choices omit a required element, include an invalid element, or use a different operation.

Question 21 - Q7(i)

Out of 200 students, 35 failed MHT-CET, 40 failed AIEEE, 40 failed IIT, 20 failed MHT-CET and AIEEE, 17 failed AIEEE and IIT, 15 failed MHT-CET and IIT, and 5 failed all three.
How many failed none?

Correct answer: 132

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 132. The other choices omit a required element, include an invalid element, or use a different operation.

Question 22 - Q7(ii)

Out of 200 students, 35 failed MHT-CET, 40 failed AIEEE, 40 failed IIT, 20 failed MHT-CET and AIEEE, 17 failed AIEEE and IIT, 15 failed MHT-CET and IIT, and 5 failed all three.
How many failed AIEEE or IIT?

Correct answer: 63

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 63. The other choices omit a required element, include an invalid element, or use a different operation.

Question 23 - Q8(i)

Of 2,000 people, 70% read Marathi newspapers, 50% read English newspapers, and 32.5% read both.
How many read at least one?

Correct answer: 1750

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 1750. The other choices omit a required element, include an invalid element, or use a different operation.

Question 24 - Q8(ii)

Of 2,000 people, 70% read Marathi newspapers, 50% read English newspapers, and 32.5% read both. How many read neither?

Correct answer: 250

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 250. The other choices omit a required element, include an invalid element, or use a different operation.

Question 25 - Q8(iii)

Of 2,000 people, 70% read Marathi newspapers, 50% read English newspapers, and 32.5% read both. How many read exactly one of the two?

Correct answer: 1100

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 1100. The other choices omit a required element, include an invalid element, or use a different operation.

Question 26 - Q9

In a hostel, 25 students take tea, 20 coffee, and 15 milk. Ten take both tea and coffee, eight take both milk and coffee, none take both tea and milk, and everyone takes at least one. How many students are in the hostel?

Correct answer: 42

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 42. The other choices omit a required element, include an invalid element, or use a different operation.

Question 27 - Q10(i)

Of 260 people with a skin disorder, 150 were exposed to chemical A, 74 to B, and 36 to both. How many were exposed to A but not B?

Correct answer: 114

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 114. The other choices omit a required element, include an invalid element, or use a different operation.

Question 28 - Q10(ii)

Of 260 people with a skin disorder, 150 were exposed to chemical A, 74 to B, and 36 to both.
How many were exposed to B but not A?

Correct answer: 38

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 38. The other choices omit a required element, include an invalid element, or use a different operation.

Question 29 - Q10(iii)

Of 260 people with a skin disorder, 150 were exposed to chemical A, 74 to B, and 36 to both.
How many were exposed to A or B?

Correct answer: 188

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives 188. The other choices omit a required element, include an invalid element, or use a different operation.

Question 30 - Q11

If $A = \{1, 2, 3\}$, which is the complete power set $P(A)$?

Options

- A. $\{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$
- B. $\{\{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$
- C. $\{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}\}$
- D. $\{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2, 3\}\}$

Correct answer: $\{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 31 - Q12(i)

Write the interval $[-3, 0]$ in set-builder form.

Options

- A. $\{x / x \in \mathbb{R}, -3 \leq x \leq 0\}$
- B. $\{x \in \mathbb{R} \mid -3 < x < 0\}$
- C. $\{x \in \mathbb{R} \mid -3 \leq x < 0\}$
- D. $\{x \in \mathbb{R} \mid -3 < x \leq 0\}$

Correct answer: $\{x / x \in \mathbb{R}, -3 \leq x \leq 0\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{x / x \in \mathbb{R}, -3 \leq x \leq 0\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 32 - Q12(ii)

Write the interval $[6, 12]$ in set-builder form.

Options

- A. $\{x / x \in \mathbb{R}, 6 \leq x \leq 12\}$
- B. $\{x \in \mathbb{R} \mid 6 < x < 12\}$
- C. $\{x \in \mathbb{R} \mid 6 \leq x < 12\}$
- D. $\{x \in \mathbb{R} \mid 6 < x \leq 12\}$

Correct answer: $\{x / x \in \mathbb{R}, 6 \leq x \leq 12\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{x / x \in \mathbb{R}, 6 \leq x \leq 12\}$. The other choices omit a required element, include an invalid element, or use a different operation.

Question 33 - Q12(iii)

Write the interval $(6, 12]$ in set-builder form.

Options

- A. $\{x / x \in \mathbb{R}, 6 < x \leq 12\}$
- B. $\{x \in \mathbb{R} \mid 6 \leq x \leq 12\}$
- C. $\{x \in \mathbb{R} \mid 6 < x < 12\}$
- D. $\{x \in \mathbb{R} \mid 6 \leq x < 12\}$

Correct answer: $\{x / x \in \mathbb{R}, 6 < x \leq 12\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{x / x \in \mathbb{R}, 6 < x \leq 12\}$. The other choices omit a required element, include an invalid element, or use a

different operation.

Question 34 - Q12(iv)

Write the interval $[-23, 5)$ in set-builder form.

Options

A. $\{x / x \in \mathbb{R}, -23 \leq x < 5\}$

B. $\{x \in \mathbb{R} \mid -23 < x < 5\}$

C. $\{x \in \mathbb{R} \mid -23 \leq x \leq 5\}$

D. $\{x \in \mathbb{R} \mid -23 < x \leq 5\}$

Correct answer: $\{x / x \in \mathbb{R}, -23 \leq x < 5\}$

Explanation and steps

Apply the definition or operation stated in the question. Evaluating the given elements or conditions gives $\{x / x \in \mathbb{R}, -23 \leq x < 5\}$. The other choices omit a required element, include an invalid element, or use a different operation.